Myositis Classification Using Deep Learning

Automated Diagnosis from Muscle Ultrasound Images

Project Overview

This project develops deep learning models to classify muscle diseases (myositis) from ultrasound images. The project addresses two classification problems:

- **Problem A**: Binary classification — Normal vs. Affected (Disease)
- **Problem B**: Multi-class classification — Dermatomyositis (DM) vs. Polymyositis (PM) vs. Inclusion Body Myositis (IBM)

Background

Inflammatory myopathies are a group of rare muscle diseases that can be challenging to diagnose. Muscle ultrasound imaging provides a non-invasive method for evaluating muscle abnormalities. This project leverages deep learning to automate the classification process, potentially improving diagnostic accuracy and speed.

Dataset

Overview

- **Total Images**: 3,214 ultrasound images
- **Source**: Patient muscle ultrasound scans

Class Distribution

Class	Label	Count
Normal	N	1,313
Dermatomyositis	D	555
Polymyositis	P	553
Inclusion Body Myositis	I	794
Total		**3,214**

Data Split

- **Training**: 70%
- **Validation**: 15%
- **Test**: 15%

Problem A: Binary Classification

Objective

Classify ultrasound images as **Normal** or **Affected** (any disease).

Model Architecture

- **Base Model**: ResNet50 (pre-trained on ImageNet)
- **Transfer Learning**: Last 100 layers unfrozen for fine-tuning
- **Classification Head**: GlobalAveragePooling2D → Dense(256, ReLU) → Dropout(0.3) → Dense(1, Sigmoid)

Training Configuration

Parameter	Value
Image Size	224 × 224
Batch Size	32
Learning Rate	1e-5
Optimizer	Adam
Loss Function	Binary Cross-Entropy
Epochs	20

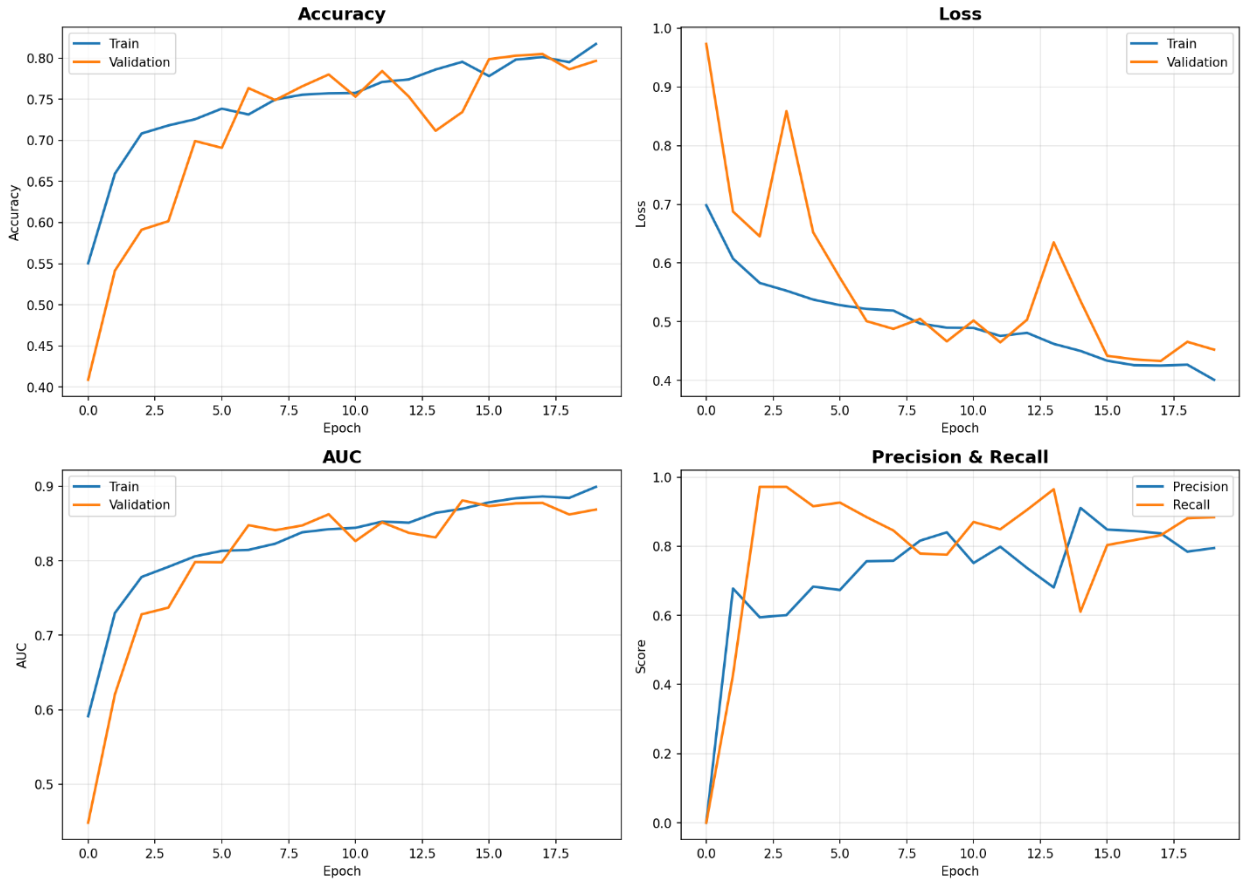
Data Augmentation

- Random horizontal flip
- Random rotation (10%)
- Random contrast (20%)
- Random zoom (10%)

Results

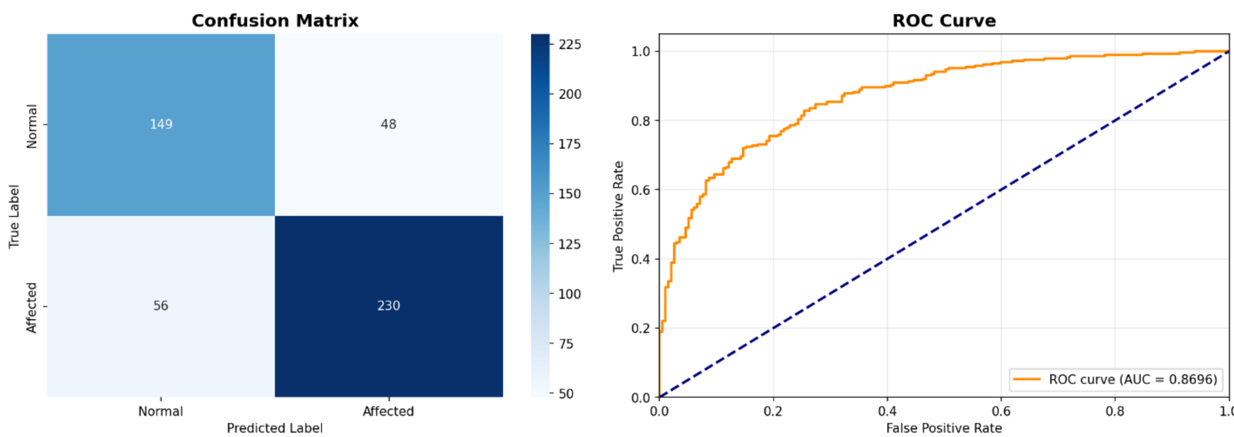
Metric	Value
Test Accuracy	**78.5%**
Test AUC	**0.87**
Test Precision	82.7%
Test Recall	80.4%
Best Validation Accuracy	80.5%

Training Curves



Problem A Training Curves

Confusion Matrix & ROC Curve



Problem A Confusion Matrix and ROC

Analysis

The binary classification model achieves strong performance with: - **78.5% accuracy** on the test set - **0.87 AUC** indicating excellent discriminative ability - Balanced precision

(82.7%) and recall (80.4%) - The ROC curve shows the model significantly outperforms random guessing

Problem B: Multi-Class Classification

Objective

Classify ultrasound images into three specific disease types: **Dermatomyositis (DM)**, **Polymyositis (PM)**, or **Inclusion Body Myositis (IBM)**.

Models Tested

Seven different architectures were evaluated: 1. ResNet50 2. ResNet101 3. EfficientNetB0 4. EfficientNetB4 5. DenseNet121 6. InceptionV3 7. Xception

Training Configuration

Parameter	Value
Image Size	224 × 224
Batch Size	32
Learning Rate	1e-5
Optimizer	Adam
Loss Function	Sparse Categorical Cross-Entropy
Max Epochs	40

Data Augmentation (Conservative)

- Random horizontal flip
- Random rotation (5%)
- Random zoom (5%)
- Random translation (4%)

Results

Model	Test Accuracy	F1-Score (Macro)	Best Val Acc	Epochs
ResNet50	**71.0%**	**0.692**	72.6%	38
Xception	64.7%	0.605	64.6%	39
ResNet101	61.5%	0.610	61.8%	35
InceptionV3	60.1%	0.552	61.4%	28
EfficientNetB4	60.1%	0.565	55.8%	35
EfficientNetB0	56.3%	0.546	53.0%	30
DenseNet121	55.9%	0.503	58.2%	35

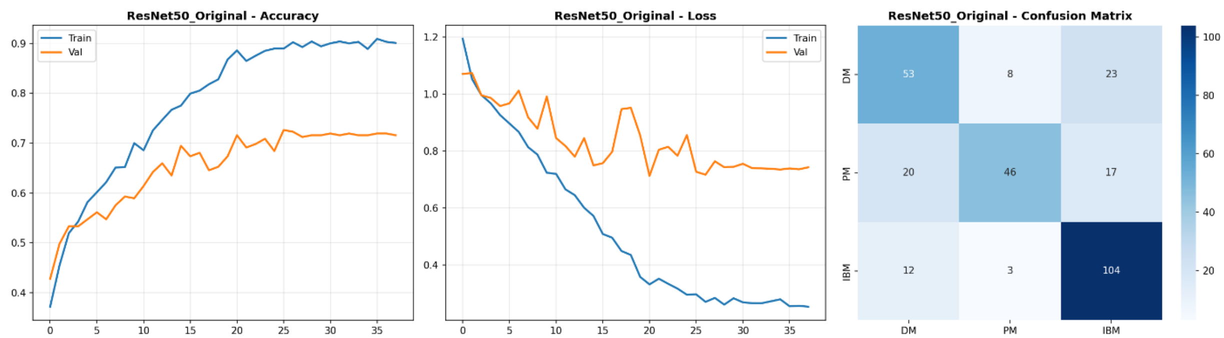
Model Comparison Table

Model Comparison - Problem B (DM vs PM vs IBM)
Learning Rate: 1e-5

model	test_accuracy	f1_macro	best_val_acc	epochs
ResNet50_Original	0.7098	0.6917	0.7263	38
Xception	0.6469	0.6055	0.6456	39
ResNet101_Original	0.6154	0.6098	0.6175	35
InceptionV3	0.6014	0.5522	0.614	28
EfficientNetB4	0.6014	0.565	0.5579	35
EfficientNetB0_Simple	0.5629	0.5456	0.5298	30
DenseNet121_Simple	0.5594	0.5032	0.5825	35

Problem B Model Comparison

Best Model: ResNet50 Results



ResNet50 Results

Analysis

- **ResNet50 performs best** with 71.0% test accuracy and 0.692 F1-score
- The confusion matrix shows IBM (I) class is classified most accurately (104 correct out of 119)
- DM and PM show more confusion with each other, which is expected given their clinical similarity
- Performance is comparable to published literature (see below)

Comparison with Literature

Reference Study

Burlina et al. (2017) - "Automated diagnosis of myositis from muscle ultrasound: Exploring the use of machine learning and deep learning methods" - Published in PLOS ONE - Achieved **74.8% accuracy** for IBM vs. DM/PM classification

Our Results vs. Literature

Metric	Burlina et al. (2017)	Our Results
Task	IBM vs. DM/PM (2-class)	DM vs. PM vs. IBM (3-class)
Test Accuracy	74.8%	71.0%
Val Accuracy	-	72.6%
Notes	Binary classification	More challenging 3-class problem

NOTE: Our 71% test accuracy (72.6% validation) on a 3-class problem is competitive with the literature's 74.8% on a simpler 2-class problem.

Technical Implementation

Software Stack

- **Python 3.x**
- **TensorFlow/Keras 2.x**
- **NumPy, Pandas, OpenCV**
- **Matplotlib, Seaborn** (visualization)
- **Scikit-learn** (metrics, class weighting)

Key Techniques

1. **Transfer Learning**: Pre-trained ImageNet weights with selective layer unfreezing
2. **Class Weighting**: Balanced weights to handle class imbalance
3. **Early Stopping**: Patience of 8-12 epochs to prevent overfitting
4. **Learning Rate Scheduling**: ReduceLROnPlateau for adaptive learning
5. **Model Checkpointing**: Save best models based on validation metrics

Code Files

- `MSK_Problem_A.ipynb`: Problem A training notebook
- `MSK_Problem_B_V2_Corrected.ipynb`: Problem B multi-model training notebook

Conclusions

Key Findings

1. **Problem A (Binary Classification)**:
   - ResNet50 achieves **78.5% accuracy** with **0.87 AUC**
   - Strong performance for screening normal vs. affected patients
2. **Problem B (Multi-Class Classification)**:
   - ResNet50 achieves best performance at **71.0% accuracy**
   - Results competitive with published literature
   - IBM classification is most accurate; DM/PM show expected overlap

Best Performing Models

Problem	Best Model	Test Accuracy	Val Accuracy	Key Metric
A (Binary)	ResNet50	78.5%	80.5%	AUC = 0.87
B (Multi-class)	ResNet50	71.0%	72.6%	F1 = 0.69

References

1. Burlina, P., et al. (2017). "Automated diagnosis of myositis from muscle ultrasound: Exploring the use of machine learning and deep learning methods." PLOS ONE. Link to Paper
2. He, K., et al. (2016). "Deep Residual Learning for Image Recognition." CVPR.
3. Tan, M., & Le, Q. V. (2019). "EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks." ICML.